

Create Department Table (Primary Key + Constraints)

```
mysql> CREATE DATABASE CollegeDB;
Query OK, 1 row affected (0.05 sec)

mysql> USE CollegeDB;
Database changed
mysql> CREATE TABLE Dept (
  ->   DeptID INT PRIMARY KEY,
  ->   DeptName VARCHAR(50) NOT NULL UNIQUE,
  ->   Location VARCHAR(50) DEFAULT 'Main Building'
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> desc Table Dept;
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | select_type | table | partitions | type | possible_keys | key | key_len | ref | rows | filtered | Extra |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | SIMPLE      | Dept  | NULL        | ALL  | NULL          | NULL | NULL    | NULL | 1    | 100.00   | NULL  |
+----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set, 1 warning (0.02 sec)
```

Constraints used:

- PRIMARY KEY → DeptID
- NOT NULL → DeptName cannot be empty
- UNIQUE → No duplicate department names
- DEFAULT → Default location

Create Students Table (Foreign Key + Constraints)

```
mysql> CREATE TABLE Students (
  -> StudentID INT PRIMARY KEY,
  -> name VARCHAR(50) NOT NULL,
  -> age INT CHECK (age >= 18),
  -> email VARCHAR(100) UNIQUE,
  -> DeptID INT,
  -> CONSTRAINT FK_Dept
  -> FOREIGN KEY (DeptID)
  -> REFERENCES Dept(DeptID)
  -> ON DELETE CASCADE
  -> ON UPDATE CASCADE
  -> );
Query OK, 0 rows affected (0.06 sec)
```

Constraints used:

- PRIMARY KEY → StudentID
- NOT NULL → name required
- CHECK → age ≥ 18
- UNIQUE → email must be unique
- FOREIGN KEY → DeptID links to Dept table
- CASCADE → auto update/delete

Insert Data

```
mysql> INSERT INTO Dept (DeptID, DeptName, Location) VALUES
    -> (1, 'Computer', 'Block A'),
    -> (2, 'Mechanical', 'Block B');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> select * from Dept;
+-----+-----+-----+
| DeptID | DeptName | Location |
+-----+-----+-----+
|      1 | Computer | Block A  |
|      2 | Mechanical | Block B  |
+-----+-----+-----+
2 rows in set (0.02 sec)

mysql> INSERT INTO Students VALUES
    -> (101, 'Pratap Londhe', 20, 'pratap172@gmail.com', 1),
    -> (102, 'Pratik Lokhande', 19, 'pratik171@gmail.com', 1),
    -> (103, 'Shubham Landge', 21, 'shubham169@gmail.com', 2);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

Test Foreign Key Constraint

```
mysql> INSERT INTO Students VALUES
    -> (104, "Pranav Pande", 21, "pranav111@gmail.com", 3);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('collegedb`.`students`,
CONSTRAINT `FK_Dept` FOREIGN KEY (`DeptID`) REFERENCES `dept` (`DeptID`) ON DELETE CASCADE ON UPDATE CASCADE)
mysql> |
```

Foreign key constraint failed because DeptID = 3 is not present in the Dept table.